Penalized Portfolio Optimization

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Abstract

This research reviews practical considerations of long-short portfolio optimization problems. It studies a few penalty forms to address the limitation of pure constraints.

1 Introduction

Modern portfolio construction is fundamentally an optimization problem. Typically, a starting portfolio $\overline{w}$ with n stocks is adjusted to a final portfolio w. The trades made in this transition incur costs and are often subject to restrictions defined by risk-aversion modeling. Position limits, factor-exposure limits, and turnover budgets are standard in quantitative equity management [10]. Models such as Markowitz's mean-variance framework [8] involve objectives and constraints that translate practical considerations into mathematical programs. Irrespective of the model, however, data utilized in these models are subject to error and could influence resulting portfolios with poor or aggressive signals. Section 2 discusses the practicalities of several core constraints and objective functions.

Each constraint imposed on a portfolio decreases the feasible set of trades leading to a common concern that constraints "drive" the solution: an optimizer may satisfy them with little consideration of the objective function. A hard cap on turnover, for example, treats a small violation and a large one identically, potentially preventing a portfolio from considering strong signals whose payoff exceeds transaction costs. Many small trades are artifacts of estimation noise or are too small to justify transaction costs. However, imposing hard thresholds to eliminate them are arbitrary and are overly simplistic.

This paper primarily studies two mechanisms for softening these rigidities. The first is a soft cap on total turnover, which allows an optimizer to exceed a specific threshold at a quadratic penalty. The second is a family of regularization penalties on individual turnovers. These induce sparsity in w and protect against noisy trading. Both families of penalties are added directly to the objective function, although some are accompanied with necessary constraints.

Certain penalties and constraints result in nonconvexity when implemented in their most primitive form, which impacts the solvability and efficiency of optimization. These restrictions are considered when they are introduced.

The regularization penalties we consider are adapted have parallels with regression. Ridge regression [6] shrinks coefficients smoothly; Lasso [12] promotes sparsity through an L_1 penalty; and the Elastic Net [14] combines the two. Hybrid models with both Lasso and Ridge have been studied in regression [9]. We will consider the practical values in these approaches with portfolio optimization, considering the Berhu (Reverse Huber) penalty [7].

Our research proceeds as follows. After establishing a background framework in Section 2, we formalize the Alpha gate as a soft total-turnover cap in Section 3. Additionally, we address concerns of non-convexity in our modeling. In section 4, we apply Elastic Net and Berhu penalties. In doing so, we derive exact convex epigraph formulations suitable for DCP solvers, and contrast the penalties' effects. Section 5, we report an empirical comparison on a proprietary dataset of trading signals. After this, we discuss parameter selection and compare the penalties.

We do not attempt a comprehensive treatment of non-convex penalties or constraints. Penalties such as SCAD [4] and MCP [13] can in principle be implemented, but their non-convexity makes

them difficult to operate at scale. We mention them as a direction for future work.

2 Implementation Framework

Modern portfolio construction involves optimizing an objective function defined over a vector of n asset positions $w \in \mathbb{R}^n$, where positive (negative) values represent long (short) positions. An example of a classic objective function is Markowitz's Mean-Variance framework:

$$\min_w \left[\frac{\lambda}{2} w' \Sigma w - r' w \right], \tag{1}$$

where r is a vector of expected returns, λ is an exogenous risk-aversion parameter, and Σ is a covariance matrix of asset returns. The design is driven by the desire to strike a balance between risks and payoffs.

Objective functions can take other forms, such as the L_2 norm of the deviation from a target vector $\tilde{w} \in \mathbb{R}^n$. A given vector $\tilde{w}$ can be a benchmark portfolio to track, or a target portfolio driven by some forecasting models. It is often desirable to mimic such a portfolio as closely as possible. One reasonable approach is to minimize this objective:

$$\min_w \|w - \tilde{w}\|_2^2.$$

A positive semi-definite covariance matrix Γ can also be incorporated to generalize the above objective function further:

$$\min_w \left[\frac{1}{2} (w - \tilde{w})' \Gamma (w - \tilde{w}) \right] \equiv \min_w \left[\frac{1}{2} w' \Gamma w - \tilde{w}' \Gamma w \right]. \tag{2}$$

Mathematically, this function is a special case of (1). Discussion of the objective will be elaborated upon in section 6.1.

A majority of portfolio construction problems can be formulated in certain quadratic forms like the above. Typical constraints include position limits, risk exposure and turnovers, etc.

Objective (1) is convex if Σ is positive semi-definite and if the constraints maintain a convex feasible set. Objective (2) acts similarly, placing both objectives within the reach of Disciplined Convex Programming (DCP). DCP problems can be easily solved with DCP solvers.

Let us focus on minimizing the family of objectives with a quadratic term $q(w)$, which captures the conventional Mean-Variance problem (1) and target tracking problem (2).

Inputs such as return estimates, risk model coefficients, and target forecasts in $\tilde{w}$ are all estimates subject to statistical errors. Thus, a problem subject to overly permissive constraints is susceptible to concentrated exposure on assets with potentially poor signals. Constraints like turnover and position limits arise from practical considerations and also serve as a form of regularization on model solutions.

On the other hand, we look to avoid overly-tight constraints. While constraints have their

purposes, each one shrinks the feasible set of potential solutions. Certain constraints may result in solutions that reflect little of the objective. Therefore, constraints should be set such that they do not "drive" the solution; the solution should be dictated in part by the objective function.

While potential payoffs are often uncertain, transaction costs are incurred upon execution. Turnover budgets and individual turnovers are often imposed, so a solution is discouraged from chasing marginal benefits with excessive transaction costs.

Linear equalities, and inequalities define hyperplanes, half-spaces respectively. These, along with second-order cones, define affine (and thus convex) sets. When feasible sets are convex, optimization is often far more efficient and accurate. Because of this, we show a preference for preserving convexity of the overall feasible region. For constraints that are non-smooth but inherently convex (such as absolute value or piecewise functions), we exercise exact epigraph reformulations with auxiliary variables. For constraints that are truly non-convex, we employ convex relaxations to approximate the feasible space while maintaining computational tractability.

2.1 Constraints

Position limits can be represented as

$$\underline{w} \leq w \leq \overline{w},$$

where constant vectors $\underline{w} \in \mathbb{R}^n$ and $\overline{w} \in \mathbb{R}^n$ are upper and lower bounds, respectively.

Similarly, limiting each individual turnover is represented by $|w - w^*| \leq t$, where constant $t \in \mathbb{R}^n$ defines individual turnover bounds, and w^* defines the starting weight vector. Turnover bounds can essentially be incorporated into position limits since w^* is a constant vector.

We define an attribute as a characteristic of an asset, which encompasses risk modeling, equity style factors, and more. Risk models are often adopted in portfolio construction to limit risk exposure, commonly in sectors and industries. Equity style factors (growth, momentum, value, etc.) are also types of attributes widely used in risk modeling. An asset can have exposures to multiple attributes. For example, although a small tech startup can have exposure in just a technology sector, a conglomerate operates in several. Attribute exposures can be negative. For example, smaller companies tend to have negative exposures to size factors.

We express n assets' exposure in m attributes with an $m \times n$ matrix $\mathbf{S}$. For example, a potential $\mathbf{S}$ for n assets across various attributes is shown in Table 1. The total portfolio exposures to the m factors are $\mathbf{S}w$. The exposure limits can be represented as

$$\underline{s} \leq \mathbf{S}w \leq \overline{s}, \tag{3}$$

where $\underline{s}$ and $\overline{s}$ are lower and upper exposure limits, respectively.

Net exposure, equal to the value of long exposure positions minus the value of short exposure positions, is a special form of attributes. It can be captured as a row of ones in $\mathbf{S}$.

Table 1: Example Factor Loading Matrix.

	Assets				
	1	**2**	**3**	...	n
Finance	0.10	0.40	0.20	...	A_n
Energy	0.13	0.20	0.30	...	B_n
Communication	0.06	0.04	0.10	...	C_n
...	...	...	...	...	...
Growth	0.70	0.30	0.20	...	D_n
Momentum	-1.0	-0.30	0.10	...	E_n
...	...	...	...	...	...

A constant upper bound $T > 0$ may be imposed on total turnover as

$$\|w^* - w\|_1 \leq T. \tag{4}$$

This introduces a total budget over the sum of individual turnovers. In practice, individual turnover limits in t are often set such that $T \leq \sum_i t_i$; otherwise, the total turnover cap will become redundant. Total turnover can be further generalized with a weighting vector $c > 0$ such that

$$c'|w^* - w| \leq T. \tag{5}$$

The vector c and T may represent some transaction cost estimates and a budget limit, respectively. When turnovers are equally weighted, c becomes e.

The L_1 constraint (4) is not directly supported by many solvers. Although convex, it is not affine, and is not differentiable at $w_i = w_i^*$. Many solvers do not natively accept L_1 norm expressions like (4). We introduce the slack vector $y \in \mathbb{R}^n$, such that

$$|w^* - w| \leq y, \quad c'y \leq T. \tag{6}$$

A feasible y exists if and only if $|w^* - w| \leq T$. Hence, (6) and (5) define the same feasible set of w.

Gross exposure, $|w\|_1$, is another measure often involved in some constraints, as it is often desirable to keep a specific amount of capital invested in a portfolio. It is common to limit the gross exposure (leverage) by an upper bound G as

$$|w\|_1 \leq G. \tag{7}$$

This is still a convex constraint like that on total turnovers. But the optimal solution may find it cheaper to settle at a gross exposure far less than G, effectively underinvesting with available capital. It is thus attractive to place a lower bound $\underline{G}$ on the gross exposure as well. When $G = \underline{G}$,

the constraint becomes an equality.

$$\|w\|_1 = G. \tag{8}$$

In long-only or short-only portfolios, absolute value signs in (8) can be trivially removed so that it becomes a convex constraint. But the presence of a lower bound or equality constraints in long-short portfolios makes the feasible set non-convex, which poses a problem for most solvers. We will address this non-convexity in section 2.3.

2.2 Karush-Kuhn-Tucker and Global Optimality Conditions

Because the portfolio construction problems considered henceforth are convex, their solutions are exact and can be calculated via DCP. Suppose a quadratic objective is bounded by affine constraints that create a convex feasible set. The Karush-Kuhn-Tucker (KKT) conditions [1] define the framework for determining optimality.

In an optimization setting with quadratic objective function, a positive semi-definite Hessian matrix ensures convexity of the contour set. When constraints are affine, the feasible set is also convex. Under such conditions, a local optimal minimum is also the global minimum.

A well-behaved convariance matrix Σ in (1) is positive semi-definite, so is Γ in (2). Almost all the constraints discussed so far are also convex with the exception of gross exposure.

2.3 Difficulties of Gross Exposure

A set $\mathcal{C} \subset \mathbb{R}^n$ is convex if and only if for any two points a and b in $\mathcal{C}$ and every $\theta \in [0, 1]$ the following is true:

$$\theta a + b(1 - \theta) \in \mathcal{C}. \tag{9}$$

We will prove that for fixed $G > 0$, the set $\mathcal{S} = \{w \in \mathbb{R}^n : \|w\|_1 = G\}$ is not convex. This argument does not restrict itself to properties of the L_1 norm. Rather, it uses two axioms every norm satisfies:

1. Absolute Homogeneity: $\|\alpha w\| = |\alpha|\|w\|$ for any scalar α.
2. Positive definiteness: $\|w\| = 0 \Leftrightarrow w = 0$.

Take any $w \in \mathcal{S}$. By homogeneity with $\alpha = -1$, we have

$$\|-w\|_1 = |-1| \cdot \|w\|_1 = \|w\|_1 \in \mathcal{S}.$$

For $\theta = \frac{1}{2}$, we have $\theta w + (1 - \theta)(-w) = 0 \notin \mathcal{S}$. Therefore, the set $\mathcal{S}$ is not convex.

The non-convexity arises from the uncertain signs of the optimal solution, which are not known prior to the solution process. We can rewrite w as $w^+ - w^-$, where both w^+ and w^- are nonnegative vectors in $\mathbb{R}^n$. This allows us to represent G as $e'(w^+ + w^-)$. The problem is that $w_i^+ w_i^-$ must equal to 0 for all i or a solver will freely choose them to hit the target G. Even Mixed-Integer

Convex Programming (MICP) will have difficulty finding an optimal solution in a timely and resource-efficient manner when n is reasonably large.

One way to overcome this is forcing the sign of $\{w_i\}$ according to some rules prior to setting up the gross exposure constraint. Let $G^+ > 0$ be the desired long exposure, and $G^- < 0$ be the negative counterpart. The total gross exposure becomes $G^+ - G^- = G$. We propose to match the sign of w to those of $\tilde{w}$ for all i such that

$$\begin{aligned}
\sum_{i \in \mathcal{W}^+} w_i &= G^+, \quad \mathcal{W}^+ = \{i | \tilde{w}_i > 0\}, \\
\sum_{i \in \mathcal{W}^-} w_i &= G^-, \quad \mathcal{W}^- = \{i | \tilde{w}_i < 0\}, \\
\sum_{i \in \mathcal{W}^0} w_i &= 0, \qquad \mathcal{W}^0 = \{i | \tilde{w}_i = 0\}.
\end{aligned}$$

The price is that the feasible set is effectively shrunken, and consequently may not yield the same global optimal solution of the original problem.

There are more ways to rid of non-convexity. Other local search methods may improve upon this approach:

- Solve the original problem with inequality (7), and use the signs of the solution vector to derive $\mathcal{W}^+$, $\mathcal{W}^-$, and $\mathcal{W}^0$. Solve the problem with the equality constraint along with the determined signs.
- After one iteration, flip the signs of securities with $|w_i|$ inside a small band ϵ to update $\mathcal{W}^+$, $\mathcal{W}^-$, and $\mathcal{W}^0$. Solve again. If there is improvement, increase ϵ and repeat. If not, stop.
- Solve the original problem with inequality (7), and scale all the weights such that $\|w\|_1 = G$. Feed the results back to iterate.

Unfortunately, these are all heuristic approaches to resolve sign uncertainty prior to the optimization. No one can guarantee the global optimum with these approaches. As long as the feasible set is not expanded like (7), the objective functions across various heuristic approaches can be compared so that they are guaranteed not to perform worse.

3 Hard and Soft Caps of Transaction Costs

Hard constraints that bound turnover and other portfolio characteristics have certain downsides. We prefer to impose them to reflect practical and risk limits, rather than dictate optimality. Violating a hard constraint by a negligible amount and by a comparatively huge amount have the exact same effect, meaning rounding errors, tiny violations, and genuine infeasibility are treated the same. Furthermore, some powerful signals may prefer to trade significantly more, either of individual securities or across the whole portfolio and the risk-adjusted payoffs far exceed the likely incurred

transaction costs. In those cases, it is desirable to permit certain violations of hard constraints. To achieve this, we introduce soft constraints.

Soft constraints allow such violations while discouraging large ones by adding a penalty to the objective. We will consider three distinct types of soft constraints: a turnover penalty, an Elastic Net, and a Berhu penalty.

The trading from w^* to w incurs transaction costs. It is thus desirable to balance the objective function and turnover to avoid paying deterministic upfront costs on insignificant or uncertain transactions. Imposing a hard budge limit on total turnover, however, handicaps portfolio construction on achieving its objective. Instead, we should consider the potential trade-offs of excessive turnovers.

In addition to a hard upper bound T on total turnover (4) or as its replacement, we can introduce a slack variable $z > 0$ to relax a constant soft bound $\tilde{T}$ such that

$$\|w - w^*\|_1 \leq \tilde{T} + z,$$

and add a penalty term to the objective function $q(\cdot)$,

$$p(z) = \alpha z^2. \tag{10}$$

This term will be referred to as an alpha gate. The exogenous parameter $\alpha > 0$ determines the cost of going above $\tilde{T}$. The variable z will be optimized along with w. The quadratic term has the benefit of having a comparable magnitude with many objective function forms of $q(w)$ such that the choice of α can be more intuitive. It is completely reasonable to replace it with $\alpha|z|$. The presence of an absolute value poses no difficult here given that $z \geq 0$.

This formulation is very intuitive. The total turnover is allowed to exceed $\tilde{T}$, but pays a penalty to do so. If turnover is less than $\tilde{T}$, no penalty is paid. The larger the value of α, the more the final turnover is discouraged from exceeding T, and vice versa. This "soft cap" is not a rigid hard bound, allowing some room to trade more when situation dictates that. When both a "hard cap" and a "soft cap" are present, we generally have $\tilde{T} < T$ in that T serves as a rigid safeguard on the total turnover.

4 Regularization of Noisy Trades

While it is a common practice to cap the total turnover $\|w-w^*\|_1$ or limit some weighted transaction costs $c'|w-w^*|$ by a budget, it is also desirable to reduce the amount of inconsequential small trades. They are either results of some computational or statistical noises, or are simply too insignificant to justify the execution or handling costs.

We wish to induce a degree of sparsity in a portfolio's turnovers, eliminating noisy trades. A common approach to encourage sparsity involves setting a certain threshold for $|w_i^* - w_i|$ below which those $\{w_i\}$ will be forced to stay at $\{w_i^*\}$, meaning no trade allowed. However, this hard rule

may be too rigid, and the threshold setting also seems rather arbitrary.

This problem is very similar to variable selection in various regression frameworks where coefficients of some variables are shrunk towards zero to yield more parsimonious models. It is thus not surprising that similar techniques can be employed to regularize $|w - w^*|$.

4.1 Elastic Net

Ridge Regression [6] softens estimated coefficients through adding an L_2 norm scaled by a multiplier. Lasso regression [12] utilizes the L_1 norm instead. The Elastic Net [14] uses both L_1 and L_2 norms simultaneously. This concept can be applied to portfolio optimization, penalizing turnovers:

$$\gamma \left[\beta \|w - w^*\|_1 + \frac{1-\beta}{2} \|w - w^*\|_2^2 \right], \tag{11}$$

where $0 \leq \beta \leq 1$ and $\gamma > 0$ are adjustable exogenous parameters.

The L_1 penalty promotes sparsity, driving small turnovers to 0. The multiplier β governs the relative aggressiveness of shrinkage. A larger β reflects a higher marginal cost of initiating a new trade. This reduces the number of small turnovers. The parameter γ controls the overall scale of this penalty term relative to the objective function $q(\cdot)$.

The L_2 penalty promotes smoothness. It prevents related trades from swinging violently with tiny input changes. Turnover among correlated assets is distributed more evenly than with only an L_1 penalty. For example, the L_2 penalty resulting from two turnovers 0.2 and 0.2 is smaller than the penalty resulting from 0.1 and 0.3, even though the total turnover is the same. Thus, the L_2 turnover results in spreading turnover more evenly across assets. The β parameters governs relative strength of penalties between the two terms.

A penalty term (11) can be added to a base quadratic objective $q(\cdot)$ along with other penalty terms such as (10). Each term in the objective is either an L_1 norm, L_2 norm, or a quadratic function. Since all scalars α, β, and γ are positive, the entire objective function maintains its convexity. The greater γ is, the more trades will be pushed towards zero.

4.2 Berhu's (Reverse Huber's) Loss

Elastic Net penalties subject portfolio weights to both L_1 and L_2 penalties, regardless of the magnitude of an asset's turnover. It has the tendency to reduce turnover indiscriminately. We introduce Berhu's (Reverse Huber) Loss, a piecewise function that contains either L_1 and L_2 penalties over non-overlapping regions.

Denote $u_i = |w_i - w_i^*|$. Frequently used in regressions, the Huber's Loss function takes the form

$$h(u_i) = \begin{cases} \frac{u_i^2}{2\epsilon} & \text{if } u_i \leq \epsilon, \\ u_i - \frac{\epsilon}{2} & \text{if } u_i > \epsilon. \end{cases} \tag{12}$$

It applies a quadratic penalty for small deviations, and a linear penalty for large deviations. This

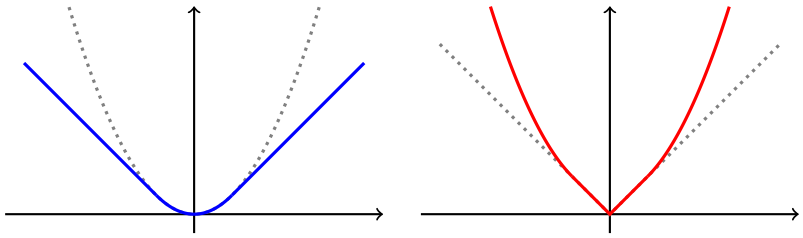


Figure 1: Huber's Loss (Left) and Berhu's Loss (Right)

mitigates the influence of extreme outliers in regression while maintaining differentiability throughout.

Berhu's Loss, also known as Reverse Huber's Loss, inverts the order of the quadratic and linear functions. Given $y_i = w_i - w_i^*$ and a threshold scalar $\epsilon > 0$, the Berhu Loss is defined as

$$b(y_i) = \begin{cases} |y_i| & \text{if } |y_i| \leq \epsilon, \\ \frac{y_i^2+\epsilon^2}{2\epsilon} & \text{if } |y_i| > \epsilon. \end{cases} \tag{13}$$

The Berhu function works similarly to an Elastic Net, but only applies either an L_1 norm or L_2 norm at a time. For minor weight adjustments where $|y_i| \leq \epsilon$, the penalty is linear, promoting sparsity. For significant weight adjustments where $|y_i| \geq \epsilon$, the penalty becomes quadratic and tend to spared trades across securities.

Intuitively, Elastic Net sharpens but does not completely eliminate the smooth edges of the contour set while the Berhu's function shrinks the areas of pointy edges but still preserve the indifferentiable vertices. Both encourage more control variables to go towards zero upon contact with the objective contour set at the optimal solution, just in slightly different fashions. Once a turnover exceeds ϵ, Berhu's function becomes quadratic while a comparable Elastic Net still maintains an L_1 term. Therefore, Berhu's functions penalize the large outliers more aggressively than the similar Elastic Net functions.

This function is convex and continuous everywhere. That said, its piecewise nature makes it difficult to implement directly into a solver, as many solvers do not accept piecewise functions natively or have a designated Berhu function. To resolve this, we'll create an epigraph formulation with auxiliary variables.

Define two non-negative auxiliary vector $u \geq 0$ and $v \geq 0$ subject to the following constraints for all i:

$$u_i + v_i \geq y_i, \quad u_i \leq \epsilon,$$

where y is defined in (6). The Berhu Loss then becomes

$$b(y) = \sum_i \left(u_i + v_i + \frac{v_i^2}{2\epsilon} \right). \tag{14}$$

When $y_i \leq \epsilon$, to minimize the penalty, the solver will set $v_i = 0$, since the quadratic penalty term $\frac{v_i^2}{2\epsilon}$

grows more rapidly than the linear term u_i, being more "expensive" in the objective function. The constraint $u_i + v_i \geq y_i$ forces $u_i = y_i$, making the penalty exactly y_i, matching the linear branch of (13). When $y_i > \epsilon$, the minimizer pushes u_i to its upper bound ϵ, resulting in v_i being non-zero $v_i = y_i - \epsilon$, while the cost becomes $\frac{y_i^2+\epsilon^2}{2\epsilon}$. This matches the value of the quadratic branch of (13).

A penalty term based on (14) can be defined as

$$b_t(y) = \gamma \sum \left(u_i + v_i + \frac{v_i^2}{2\epsilon} \right) = \gamma \sum_i \left[y_i + \frac{1}{2\epsilon} \max[y_i - \epsilon, 0]^2 \right], \tag{15}$$

where $\gamma > 0$ is an exogenous scale coefficient that determines the scale of this term relative to the objective function $q(\cdot)$ and other penalty terms present. All terms in the objective function introduced so far are either quadratic or linear, so the objective function remains convex.

Threshold ϵ is the inflection point where the penalty transitions from linear to quadratic. A small ϵ means that the optimizer switches to quadratic penalties at very small turnover values, likely resulting in fewer turnovers reduced to exactly 0. A larger ϵ means the penalty remains linear for a wider range of turnovers, penalizing large trades less aggressively. The choice of ϵ is highly sensitive to the data scales, and the nature of the data.

5 Numerical Experiments

We base our numerical experiments on a proprietary dataset provided by a hedge fund manager. The dataset contains 10 years of daily trading signals for the most liquid US stocks. There are three signal series intended as target portfolio weights, varying in natural turnovers. They were generated from some quantitative models that were not concerned with various portfolio construction considerations. Our objective is to implement long-short portfolios that replicate these target weights as closely as possible subject to various constraints. We focus on a special form of loss function (2) with Γ being an identity matrix,

$$\min_w \|w - \tilde{w}\|_2^2,$$

where $\tilde{w}$ represents one of the model target weight vector.

Let $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ represent the three signal series from this dataset. Table 2 contains their summary statistics. All the portfolio weights are net of their means and then normalized by the gross exposures. Because $\mathcal{D}_1$ has the highest natural turnover, the turnover constraints are most likely to bind for this series; we therefore use $\mathcal{D}_1$ as the main diagnostic series in the parameter studies below.

5.1 Experimental Design and Metric Definitions

To maintain comparability, we define standard constraints for all the experiments:

- $|w\|_1 = 2$.

Table 2: Summary Statistics of the Raw Dataset (01/04/2016 – 12/31/2025)

	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
Daily Average of Number of Qualified Stocks	2,201	2,201	2,201
Daily Average of Meaningful Positions ($> 1^{-8}$)	1,102	1,103	1,103
Daily Average of Total Turnover (%)	131.7	91.83	76.75
Daily Average of Position Standard Deviations	.154	0.159	0.165
Daily Average of Max Position Sizes (%)	1.400	1.558	1.810
Daily Average of Max Change Magnitudes (%)	1.117	0.957	0.938

- $-0.04 \leq w \leq 0.04$.
- $|w - w^*| \leq 0.02$.
- $\|w^* - w\|_1 \leq 0.7$.
- $-10^{-3} \leq \mathbf{S}w \leq 10^{-3}$, where $\mathbf{S}$ is a $30 \times n$ matrix containing various sector exposures.
- $e'w = 0$.

These constraints are quite normal in actual trading environments.

We ran a baseline iteration with these standard constraints for signal series $\mathcal{D}_i$ where $i = 1, 2, 3$. For each experiment and signal series, we computed few metric series over time k:

- $o_k^{(m,i)}$: total turnover;
- $l_k^{(m,i)}$: realized loss $\|w_k - \tilde{w}_k\|_2^2$;
- $c_k^{(m,i)}$: correlation between optimized w and target $\tilde{w}$;
- $z_k^{(m,i)}$: portion of optimized trades with magnitude below 10^{-8}.

For the experiments involving the Elastic Net penalty (11), let $e(\beta, \gamma)$ be the experiment with parameter (β, γ). We represent the corresponding metric series as $\{l_k^{(e,i)}(\beta = \cdot, \gamma = \cdot)\}$ (without the penalty term), $\{c_k^{(e,i)}(\beta, \gamma)\}$, and $\{z_k^{(e,i)}(\beta, \gamma)\}$, respectively. Similarly for Berhu penalty (15), we have $\{l_k^{(b,i)}(\epsilon, \gamma)\}$, $\{c_k^{(b,i)}(\epsilon, \gamma)\}$, and $\{z_k^{(b,i)}(\epsilon, \gamma)\}$, respectively. For iterations involving the Alpha penalty, we have $\{l_k^{(a,i)}(\alpha)\}$, $\{c_k^{(a,i)}(\alpha)\}$, and $\{o_k^{(a,i)}(\alpha)\}$.

Based on these metric series, we define the following averages over n days:

$$L^{(e,i)}(\beta,\gamma) = \frac{1}{n}\sum_{k=1}^{n}\Big[l_k^{(e,i)}(\beta,\gamma) \div l_k^{(0,i)}(\beta,\gamma)\Big],$$

$$C^{(e,i)}(\beta,\gamma) = \frac{1}{n}\sum_{k=1}^{n}\Big[c_k^{(e,i)}(\beta,\gamma) - c_k^{(0,i)}(\beta,\gamma)\Big],$$

$$Z^{(e,i)}(\beta,\gamma) = \frac{1}{n}\sum_{k=1}^{n}\Big[z_k^{(e,i)}(\beta,\gamma) - z_k^{(0,i)}(\beta,\gamma)\Big],$$

$$L^{(b,i)}(\epsilon,\gamma) = \frac{1}{n}\sum_{k=1}^{n}\Big[l_k^{(b,i)}(\epsilon,\gamma) \div l_k^{(0,i)}(\epsilon,\gamma)\Big],$$

$$C^{(b,i)}(\epsilon,\gamma) = \frac{1}{n}\sum_{k=1}^{n}\Big[c_k^{(b,i)}(\epsilon,\gamma) - c_k^{(0,i)}(\epsilon,\gamma)\Big],$$

$$Z^{(b,i)}(\epsilon,\gamma) = \frac{1}{n}\sum_{k=1}^{n}\Big[z_k^{(b,i)}(\epsilon,\gamma) - z_k^{(0,i)}(\epsilon,\gamma)\Big],$$

$$L^{(a,i)}(\alpha) = \frac{1}{n}\sum_{k=1}^{n}\Big[l_k^{(a,i)}(\alpha) \div l_t^{(0,i)}(\alpha)\Big],$$

$$C^{(a,i)}(\alpha) = \frac{1}{n}\sum_{k=1}^{n}\Big[c_k^{(a,i)}(\alpha) - c_k^{(0,i)}(\alpha)\Big],$$

$$O^{(a,i)}(\alpha) = \frac{1}{n}\sum_{k=1}^{n}\Big[o_k^{(a,i)}(\alpha) - o_k^{(0,i)}(\alpha)\Big].$$

Table 3: Baseline Results

	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
Average absolute position size (%)	0.035	0.035	0.035
Average $\{o_k^{(0,i)}\}$	≈ 0.700	≈ 0.700	≈ 0.700
Average $\{c_k^{(0,i)}\}$	0.961	0.984	0.988
Average $\{z_k^{(0,i)}\}$	0.452	0.250	0.162
Average $\{l_k^{(0,i)}\}(\times 10^{-4})$	4.068	1.687	1.319

The baseline results are presented in table 3. Although we iterated once with all the three signal series, we will focus on the speed $\mathcal{D}_1$ where total turnover constraints are more likely to be binding.

5.2 Impacts of Alpha Gating

Graphs with the Alpha penalty are displayed with metrics $L^{(a,1)}(\alpha)$, $C^{(a,1)}(\alpha)$, and $O^{(a,1)}(\alpha)$. For all iterations with the Alpha penalty, we set threshold $\tilde{T} = 0.4$ past which the penalty is incurred. All three graphs act as expected. C and O seem inversely related with α, while L is directly related.

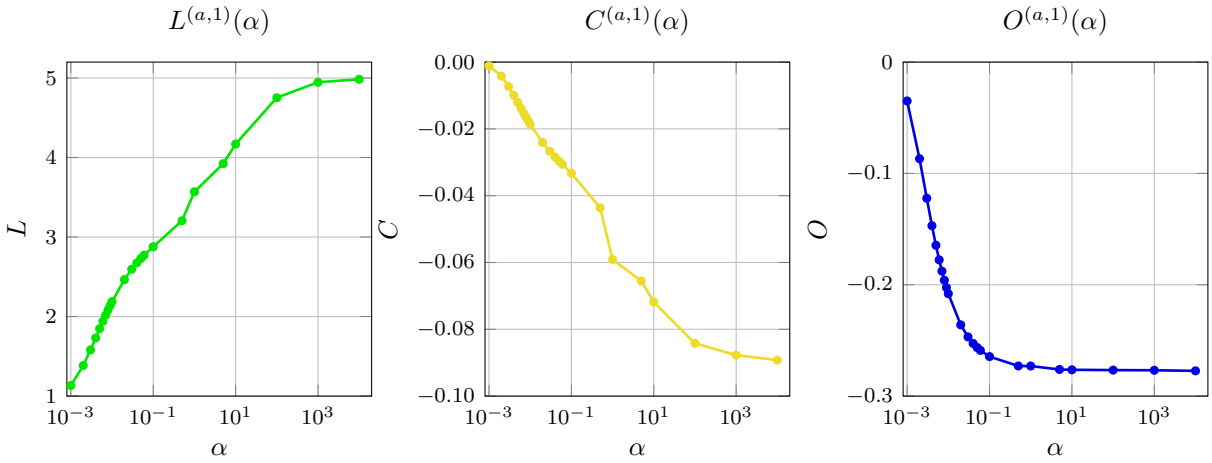


Figure 2: Alpha penalty optimization across metrics.

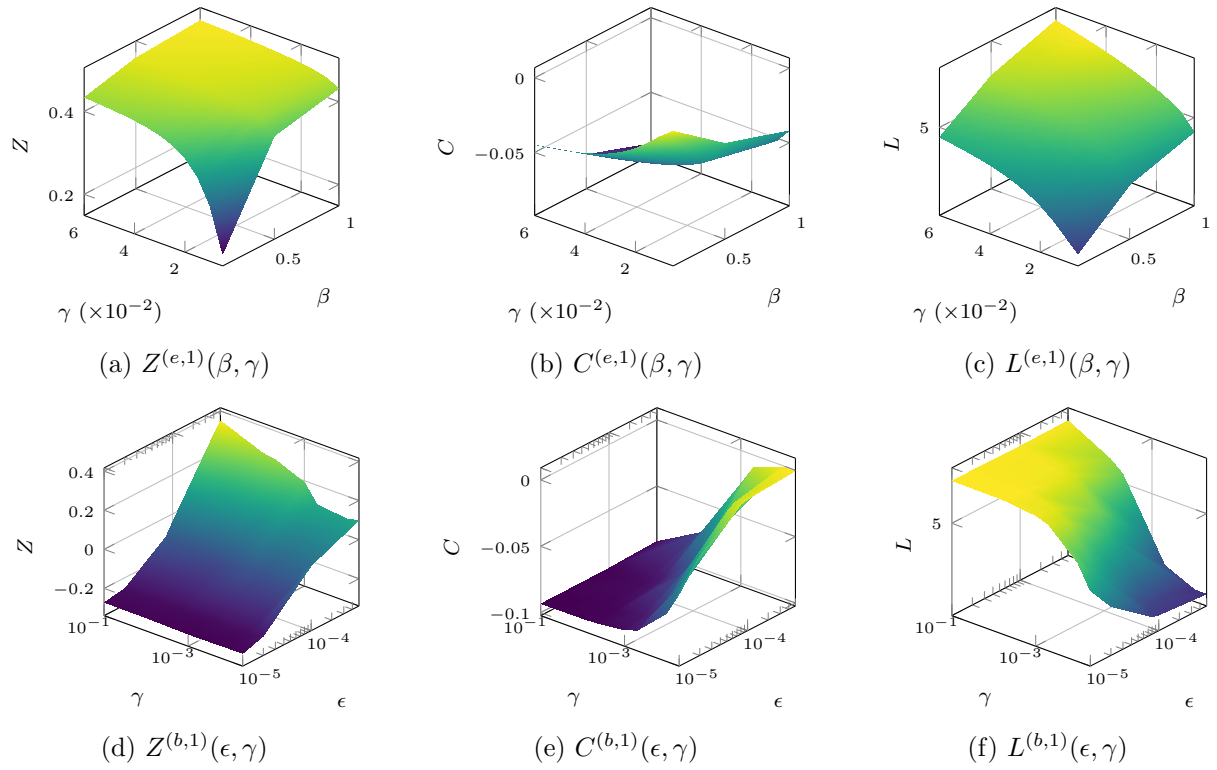
Figure 2 shows $L^{(a,1)}(\alpha)$ rises sharply at first and then plateaus around $\alpha = 10$. This direct relationship is to be expected: the coefficient amplifies the effect of the penalty, increasing the loss. $C^{(a,1)}(\alpha)$ drops then plateaus when $\alpha = 10$, which is also expected since correlation would decrease as the penalty increased. $O^{(a,1)}(\alpha)$ decreases precipitously as α increases, then later asymptotes at a value of around -0.28. This implies the realized turnover approaches 0.42. Considering that $T = 0.7$ and $\tilde{T} = 0.4$ and that constraints are binding, the penalty successfully acts as a soft barrier. The values of α we used in iterations were largely concentrated between 10^{-3} and 10^{-1} to observe the range with more rapid change in turnover. The asymptotes in the three graphs reveal that increasing α beyond ≈ 0.1 results in close to no reduction in turnover $O^{(a,1)}(\alpha)$, but continues to slowly diminish correlation and increase loss. Past this point, there will no longer be significant value in increasing α if the goal is controlling turnover.

5.3 Selection of Berhu and Elastic Net Parameters

The values of ϵ and β in Berhu and Elastic Net penalties respectively have disparate effects on optimized portfolios. Elastic Net is similar to dialing two switches. The L_1 penalty discourages small trades and the L_2 penalty, which almost always is active, discourages sparsity. In practice, the penalty applies both effects at each trade size. Meanwhile, Berhu is a pure L_1 penalty for small trades and a pure L_2 penalty for large ones.

We should select exogenous parameters in relation to the loss function. The relative magnitude of the penalty to the loss will define the strength of the downward or upward forces on weights.

For elastic graphs in Figure 3, a β range of $(0, 1]$ was selected to consider balancing the penalties, while γ was selected within a range of $[5 \cdot 10^{-3}, 6 \cdot 10^{-2}]$. For Berhu graphs, ϵ was within a range of $(10^{-5}, 5 \cdot 10^{-4})$ was selected, while γ was selected within a range of $[5 \cdot 10^{-5}, 10^{-1}]$.



(a) $Z^{(e,1)}(\beta, \gamma)$ (b) $C^{(e,1)}(\beta, \gamma)$ (c) $L^{(e,1)}(\beta, \gamma)$

(d) $Z^{(b,1)}(\epsilon, \gamma)$ (e) $C^{(b,1)}(\epsilon, \gamma)$ (f) $L^{(b,1)}(\epsilon, \gamma)$

Figure 3: Comparison of Elastic Net and Berhu optimization across metrics.

5.4 Impacts of Berhu and Elastic Net Penalties

Graphs for various data points in both Berhu and Elastic Net iterations are displayed in Figure 3.

If the target portfolio weights are derived with some stable modeling process, a set of penalty parameters can be evaluated based on its consistency over a sample period. For example, a set of regularization parameters is ideal when they are the least aggressive while shrinking just enough number of trades to zero consistently throughout the sample period. An alpha gate is ideal when it prevents the hard total turnover from being hit most time.

In an objective function with an Elastic Net penalty, the marginal cost of a trade y is $\gamma[\beta + (1-\beta)y]$. With a Berhu penalty, it becomes γ for $y \leq \epsilon$, $\frac{\gamma y}{\epsilon}$ beyond. In Elastic Nets, the threshold under which smaller trades are penalized hinges on the value of $\gamma\beta$, as the L_1 penalty gains this coefficient. Meanwile, the Berhu penalty sets a specific threshold in ϵ. This difference is reflected in Figure 3. In the Elastic Net graphs, the value on the $z-$axis is contingent on both β and γ. In the Berhu graphs, the z value is impacted predominantly by one or the other, although both do influence it.

6 Concluding Remarks

The penalized portfolio optimization techniques studied in this paper address two deficiencies of the conventional approach. The first one concerns with rigid constraints that sometimes end up

driving the solution. The alpha gate introduced in Section 3 directly incorporate violation of a soft threshold in an objective function. It can be viewed as a flexible constraint that can be stretched to some degree. The second one serves as a shrinkage mechanism, similar to variable selection in regression settings. The regularization methods introduced in Section 4 aims to eliminate noisy trades.

A main concern with penalty terms is that they tend to add noises to original objective functions. This is particularly true in the case of regularization methods. One way to mitigate the bias is to follow a common practice in regression studies. After identifying the relevant features, a regression is run with only the selected features. In the portfolio optimization, the noise trades that fall below a tiny threshold from a penalized optimization can be forced to zero and another optimization is then performed without such a penalty term. This ensures that the optimal solution is not biased by regularization biases.

Another way to mitigate penalty biases takes form of non-convex penalty functions. The SCAD penalty by [4] and the MCP penalty by [13] can be applied to $|w - w^*|$ to penalize turnovers. Like elastic net and Berhu, both penalty functions are non-differentiable at zero, allowing for sparsity. But unlike those convex penalities, they are concave elsewhere. Consequently, the marginal penalty gradually approaches zero as $\|w - w^*\|_1$ increases and the total penalty stops increasing. Effectively, such penalty forms set a ceiling on the penalty biases. However, non-convexity of these functions makes it very difficult to optimize.

6.1 Expansion: Huber's Loss for Target Tracking

Up to this point, the base objective family $q(\cdot)$ we used was a quadratic function along with some linear or quadratic penalty terms represented by $p(\cdot)$, an example of which is (2). A target vector $\tilde{w}$ may represent a benchmark portfolio or a model driven portfolio. It is often desirable to form portfolios as close as possible to such a given portfolio subject to various constraints.

The Huber's Loss function (12) is particularly interesting as an alternative objective function. For an individual tracking error $x_i = w_i - \tilde{w}_i$, and a threshold parameter δ, the Huber's Loss function can be written as

$$h(x_i) = \begin{cases} \frac{x_i^2}{2\delta} & \text{if } x_i \leq \delta, \\ x_i - \frac{\delta}{2} & \text{if } x_i > \delta. \end{cases}$$

Such a loss function dampens the larger points beyond the threshold δ. It reduces impact of large deviations from the target portfolio and may indirectly lower the tendency of high turnovers.

Similar to (14), this can be reformulated to a single equation using auxiliary variables u_i and v_i, then scaled by a factor λ:

$$h(x_i) = \lambda \left(\frac{u_i^2}{2\delta} + v_i \right), \tag{16}$$

where

$$\begin{aligned} u_i + v_i &\geq x_i, \\ 0 \leq u_i &\leq \delta, \\ v_i &\geq 0. \end{aligned}$$

CVXPY [3] contains designated Huber functions that allows direct specification without slack vectors u and v. Behind the scene, however, it carries out the similar conversions from Huber functions.

The loss function (16) can be made the objective function, along with other penalties such as turnover gating, Elastic Net, or Berhu's Loss. In portfolio optimization, minimizing the squared tracking error is sensitive to outliers. The Huber's Loss based objective function penalizes larger $|w - \tilde{w}|$ less than the quadratic one. The value of λ dictates how much the solver discourages responses to large outliers in $\tilde{w}$ above δ.

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